



[\(https://www.aetna.com/\)](https://www.aetna.com/)

---

# Diabetes Tests, Programs and Supplies

[Clinical Policy Bulletins](#) | [Medical Clinical Policy Bulletins](#)

---

**Number: 0070**

## Table Of Contents

[Policy](#)

[Applicable CPT / HCPCS / ICD-10 Codes](#)

[Background](#)

[References](#)

## Policy History

[Last Review](#)  02/20/2026

Effective: 10/20/1995

Next Review: 01/14/2027

---

[Review History](#) 

---

[Definitions](#) 

## Additional Information

[Clinical Policy Bulletin Notes](#) 

# Policy

## Scope of Policy

This Clinical Policy Bulletin addresses diabetes tests, programs and supplies.

**Note:** For Statement of Medical Necessity (SMN) precertification forms, see [Specialty Pharmacy Precertification \(https://www.aetna.com/health-care-professionals/health-care-professional-forms.html\)](https://www.aetna.com/health-care-professionals/health-care-professional-forms.html).

### I. Medical Necessity

Aetna considers the following medically necessary for members with diabetes:

#### A. *Diabetes Self-Care Programs*

Outpatient medical self-care programs when such programs meet the following criteria:

1. The program consists of services of recognized healthcare professionals (e.g., physicians, registered dietitians, registered nurses, registered pharmacists);  
*and*
2. The program is designed to educate the member about medically necessary diabetes self-care; *and*
3. The program is ordered by the physician treating the member's diabetes and includes a statement signed by the physician that the service is needed.

#### B. *Diabetic Supplies*

The following diabetic supplies:

1. Alcohol swabs;
2. Blood glucose monitors;
3. Blood glucose test strips;
4. Control solutions;
5. Insulin pens;
6. Lancets;
7. Needles and syringes for insulin administration; *and*
8. Urine test tablets/strips.

*C. Glutamic Acid Decarboxylase (GAD) Autoantibodies*

1. Measurement of autoantibodies to GAD for distinguishing type 1 from type 2 diabetes when the clinical history is ambiguous and the results of testing will influence patient management (See [CPB 1022 - Teplizumab-mzwv \(Tzielid\) \(../1000\\_1099/1022.html\)](#));
2. Measurement of anti-GAD antibodies in diagnosing stiff-person syndrome (See [CPB 0340 - Antibody Tests for Neurologic Diseases \(../300\\_399/0340.html\)](#)).

Anti-GAD antibody measurement is considered experimental, investigational, or unproven for predicting the onset of diabetes and for all other indications.

*D. Jet Injectors*

Jet injectors (e.g., Vita-Jet II, Advanta Jet, Freedom Jet, Medi-Jector EZ, Biojector 2000) as durable medical equipment (DME) when the member or the member's caregiver is physically unable to use a conventional needle-syringe.

The use of jet injectors for other reasons is considered a matter of preference and convenience.

E. *Continuous Glucose Monitoring (CGM) Devices*

1. Short-term (72 hours to 1 week) *diagnostic use* of continuous glucose monitoring (CGM) devices for the following indications:

a. For members with diabetes who have *either* of the following problems in controlling blood glucose level, unresponsive to conventional insulin dose adjustment:

i. Hypoglycemia unawareness; *or*

ii. Repeated hypoglycemia (less than 50 mg/dL) and hyperglycemia (greater than 150 mg/dL) at the same time each day; *or*

b. To diagnose primary islet cell hypertrophy (nesidioblastosis) or persistent hyperinsulinemic hypoglycemia of infancy (PHHI) (congenital hypoglycemia) in members with symptoms suggestive of recurrent hypoglycemia. For short-term (72 hours to 1 week) diagnostic use, no more than 2 continuous glucose monitoring periods within a 12-month period;

2. Long-term (greater than 1 week) therapeutic use of continuous glucose monitoring (CGM) devices (e.g., Dexcom, Eversense, Freestyle Libre, Guardian) for the following indications:

a. Criteria for Initial Approval



i. Member has a diagnosis of diabetes mellitus (type 1 or type 2) and meets *both* of the following:

a. The member is using an intensive insulin regimen (defined as multiple daily injections [i.e., 3 or more injections per day] or insulin pump therapy); *and*

b. The member meets *one* of the following criteria:

- The member is less than 18 years of age; *or*
- The member is not meeting glycemic targets; *or*
- The member is experiencing hypoglycemia (including hypoglycemia unawareness); *or*

ii. Member has a diagnosis of glycogen storage disease;

b. Continuation of Therapy

i. The member has a diagnosis of diabetes mellitus and meets *all* of the following criteria:

a. The member is using an intensive insulin regimen (defined as multiple daily injections [i.e., 3 or more injections per day] or insulin pump therapy); *and*

b. The member meets *one* of the following:

- The member has experienced improved glycemic control or decreased hypoglycemia episodes while using a CGM; *or*
- The member is being assessed every 6 months by the prescriber for adherence to their CGM regimen and diabetes treatment plan;

*or*

- ii. All members (including new members) with diagnosis of glycogen storage disease meets all initial criteria.

Long-term therapeutic use of a CGM device is considered experimental, investigational, or unproven for all other indications, including the following (not an all-inclusive list), because there is insufficient evidence of the clinical benefits of this approach for these indications:

- a. Gestational diabetes
- b. Member with type 2 diabetes not using intensive insulin regimens;
- c. Monitoring blood glucose in non-diabetic members following gastric bypass surgery
- d. Neonatal hypoglycemia
- e. Nesidioblastosis (primary islet cell hypertrophy).

### *3. Artificial Pancreas Device Systems*

For artificial pancreas device system that incorporate a continuous glucose monitor and insulin infusion pump, see [CPB 0161 - Infusion Pumps \(../100\\_199/0161.html\)](#)

For Trina Health artificial pancreas treatment, see [CPB 0742 - Intermittent Intravenous Insulin Therapy \(../700\\_799/0742.html\)](#).

### *4. Blood Glucose Meters for Members with Visual Impairment*

Reflectance meters with an electronic voice, automatic timers, and specially designed arrangements of supplies and materials to allow the visually impaired to use the equipment without assistance as DME only for legally blind (best corrected visual acuity less than 20/200) members with diabetes.

5. *Blood Glucose Monitors with Integrated Lancing/Blood Sample*

Blood glucose monitors with integrated lancing/blood sample as DME in members with diabetes who meet *either* of the following criteria:

- a. Members who are legally blind (best corrected visual acuity less than 20/200); *or*
- b. Members with impairment of manual dexterity severe enough to require the use of this special monitoring system.

6. *Alternate Site Blood Glucose Monitors*

Alternate site blood glucose monitors as DME for the following members with diabetes, when an alternate site blood glucose monitor is recommended by their physician:

- a. Children below age of 12 years; *or*
- b. Members who have used conventional blood glucose meters for at least 1 month (more than 30 days) and who have been non-compliant with blood glucose testing because of pain sensitivity or heavily callused fingertips.

Alternate site blood glucose monitors have no proven value over standard blood glucose monitors for other indications.

#### *7. Disposable Blood Glucose Monitor*

A disposable blood glucose monitor (e.g., the ReliOn NewTek [Hypoguard USA, Inc., Edina, MN]) is an acceptable alternative to a standard blood glucose monitor.

#### *8. Insulin Infusion Pumps*

For insulin infusion pumps, please see [CPB 0161 - Infusion Pumps \(./100.199/0161.html\)](http://100.199.0161.html).

#### *9. Flash Glucose Monitoring Systems*

Intermittently scanned "flash" continuous glucose monitoring systems (FreeStyle Libre Flash Glucose Monitoring System) are an equally acceptable alternative to other continuous glucose monitoring systems for medically necessary indications.

#### *10. Implantable Glucose Sensors*

Continuous glucose monitors with implantable glucose sensors (e.g., the Eversense E3 implantable CGM sensor) are an equally acceptable alternative to standard continuous glucose monitors for medically necessary indications.

#### *11. I-Port*

Aetna considers the I-Port Injection Port (Patton Medical) a non-covered convenience item.

#### 12. *Combinational Items*

Aetna considers combination devices that include a home blood glucose monitor combined with a blood pressure monitor, cholesterol screening analyzer, or other devices (e.g., cellular telephone) not specifically indicated for the management of diabetes mellitus as not medically necessary convenience items.

#### 13. *Hypoglycemic Wristband Alarms*

Aetna considers hypoglycemic wristband alarms (e.g., Sleep Sentry) a noncovered convenience item.

#### 14. *Cellular Glucometry*

Aetna considers a feature that allows wireless transmission of blood glucose test results (cellular-enabled glucometer) as an integral part of the glucometer and not separately reimbursed.

### F. Experimental, Investigational, or Unproven

The following interventions are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

#### 1. *Lasette™ Laser Blood Glucose Monitoring Device*

The Lasette laser blood glucose monitoring device (Cell Robotics International Inc., Albuquerque, NM) uses a laser instead of a lancet to perforate the skin to obtain a blood sample for glucose measurement. There is insufficient evidence in the peer-reviewed medical literature that laser skin perforation offers clinically significant advantages over standard lancets.

## *2. Glycated Serum Proteins (GSP)*

The clinical utility of monitoring glycated serum proteins with devices to measure glycated serum proteins (fructosamine) (e.g., Duet™ Glucose Control System by LXN Corporation) has not been established.

## *3. PreDX Test*

The PreDx Test has inadequate evidence in the published peer-reviewed clinical literature regarding its effectiveness.

## *4. GlucoWatch® Biographer Monitor*

The GlucoWatch Biographer (Cygnus Inc, Redwood City, CA.), a glucose meter that is worn on the wrist.

## *5. Home Glycated Hemoglobin Monitors*

For home glycated hemoglobin (HbA1c or A1C) monitors (e.g., A1cNow Diabetes Monitor, Metrika Inc., Sunnyvale, CA) there are no prospective clinical studies demonstrating improvements in compliance or other clinically significant

benefits of home A1C testing over laboratory A1C testing. Individual-case exceptions to this policy may be made upon medical review for members who are unable to access laboratory A1C testing.

#### *6. Diabetes Management Software*

Aetna considers mobile application software (e.g., BlueStar, d-Nav) for self-management of diabetes experimental, investigational, or unproven because its effectiveness has not been established.

**Note:** Aetna considers computer software for analyzing blood glucose monitor test results as an integral part of a blood glucose monitor and not separately reimbursed. In addition, software or hardware required for downloading data from a blood glucose monitor to a computer are considered an integral part of the blood glucose monitor and not separately reimbursed.

For mobile application software for self-management of diabetes, see [CPB 0999 - Prescription Digital Therapeutics \(./900\\_999/0999.html\)](#).

#### *7. Personal Digital Assistant-Based Blood Glucose Monitor*

A personal digital assistant-based blood glucose monitoring device (e.g., TheraSense FreeStyle Tracker, Accu-Check Advantage Module) and module have not been shown in published clinical studies to improve clinical outcomes over standard blood glucose monitors. **Note:** A personal digital assistant (PDA) does not meet Aetna's definition of covered DME in that the PDA can be used in the absence of illness or injury.

#### *8. Infrared Thermometer Device*

There is insufficient evidence for the effectiveness of an infrared thermometer device (e.g., TempTouch) for the intermittent measurement and monitoring of skin surface temperature in reducing the risk for diabetic foot ulceration.

#### *9. Measurement of Advanced Glycation End Products by Skin Autofluorescence*

There is insufficient evidence of the effectiveness of measurement of advanced glycation end products by skin autofluorescence compared to the oral glucose tolerance test.

#### *10. Remote Glucose Monitoring*

There is insufficient published evidence of the impact of remote glucose monitoring on clinical outcomes by remote wireless glucose monitoring devices (e.g., mySentry) for managing members with diabetes. Aetna does not cover an attachment to allow wireless transmission from a continuous glucose monitor to a smart phone or computer (e.g., MiniMed Connect) because it is considered a convenience feature. Aetna provides no additional reimbursement for a wireless transmission feature that is integrated into a continuous glucose monitor (e.g., Dexcom SHARE) because it is considered a convenience feature.

#### *11. Cellular Activation Therapy*

Cellular activation therapy by means of the Bionica Microdose infusion pump (Diabetic Innovations, Franklin lakes, NJ).

#### *12. Skin Microvascular Flow-Motion*

Skin microvascular flow-motion for the management of diabetes mellitus.



### 13. Kihealth Diabetes Risk Test

#### G. Policy Limitations and Exclusions

**Note:** Except for Medicare plans and where coverage is mandated by state law, generally coverage for diabetic supplies would be provided under a pharmacy rider and not as part of medical coverage. Certain diabetic supplies may also be covered under the medical plan if no pharmacy or diabetic supplies rider is available. Please check plan benefits.

**Note:** Coverage of diabetic supplies varies by medical and pharmacy plan. Please check plan documents for details.

#### H. Related Policies

- [CPB 0049 - Nutritional Counseling \(../1\\_99/0049.html\)](#)
- [CPB 0161 - Infusion Pumps \(../100\\_199/0161.html\)](#)
- [CPB 0587 - Pancreas Kidney Transplantation \(../500\\_599/0587.html\)](#)  
[CPB 0601 - Pancreas Transplantation Alone \(PTA\) and Islet Cell Transplantation \(../600\\_699/0601.html\)](#)
- [CPB 0742 - Intermittent Intravenous Insulin Therapy \(../700\\_799/0742.html\)](#)
- [CPB 0999 - Prescription Digital Therapeutics \(../900\\_999/0999.html\)](#)
- [CPB 1022 - Teplizumab-mzwv \(Tzield\) \(../1000\\_1099/1022.html\)](#) - discusses glutamic acid decarboxylase 65 (GAD) autoantibodies, insulin autoantibody (IAA), insulinoma-associated antigen 2 autoantibody (IA-2A), zinc transporter 8 autoantibody (ZnT8A), and islet cell autoantibody (ICA) testing.

## CPT Codes / HCPCS Codes / ICD-10 Codes

| Code                                                     | Code Description                                                                                                                                                                                             |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPT codes covered if selection criteria are met:         |                                                                                                                                                                                                              |
| 0403T                                                    | Preventive behavior change, intensive program of prevention of diabetes using a standardized diabetes prevention program curriculum, provided to individuals in a group setting, minimum 60 minutes, per day |
| 82947                                                    | Glucose; quantitative, blood (except reagent strip)                                                                                                                                                          |
| 82948                                                    | blood, reagent strip                                                                                                                                                                                         |
| 82950                                                    | post glucose dose (includes glucose)                                                                                                                                                                         |
| 82962                                                    | Glucose, blood by glucose monitoring device(s) cleared by the FDA specifically for home use                                                                                                                  |
| 83519                                                    | Immunoassay, analyte, quantitative; by radiopharmaceutical technique (eg, RIA)                                                                                                                               |
| 86341                                                    | Islet cell antibody                                                                                                                                                                                          |
| CPT codes not covered for indications listed in the CPB: |                                                                                                                                                                                                              |
| 0602U                                                    | Kihealth Diabetes Risk Test                                                                                                                                                                                  |
| 0740T                                                    | Remote autonomous algorithm-based recommendation system for insulin dose calculation and titration; initial set-up and patient education                                                                     |
| 0741T                                                    | Remote autonomous algorithm-based recommendation system for insulin dose calculation and titration; provision of software, data collection, transmission, and storage, each 30 days                          |

| Code                                                      | Code Description                                                                                                                                                                                                                                      |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 81506                                                     | Endocrinology (type 2 diabetes), biochemical assays of seven analytes (glucose, HbA1c, insulin, hs-CRP, adiponectin, ferritin, interleukin 2-receptor alpha), utilizing serum or plasma, algorithm reporting a risk score [PreDx Diabetes Risk Score] |
| <b>Other CPT codes related to the CPB:</b>                |                                                                                                                                                                                                                                                       |
| 83036                                                     | Hemoglobin; glycosylated (A1C)                                                                                                                                                                                                                        |
| 83037                                                     | glycosylated (A1C) by device cleared by FDA for home use                                                                                                                                                                                              |
| 97802                                                     | Medical nutrition therapy; initial assessment and intervention, individual, face-to-face with the patient, each 15 minutes                                                                                                                            |
| 97803                                                     | re-assessment and intervention, individual, face-to-face with the patient, each 15 minutes                                                                                                                                                            |
| 97804                                                     | group (2 or more individual(s)), each 30 minutes                                                                                                                                                                                                      |
| <b>HCPCS codes covered if selection criteria are met:</b> |                                                                                                                                                                                                                                                       |
| A4206                                                     | Syringe with needle, sterile 1 cc or less, each                                                                                                                                                                                                       |
| A4207                                                     | Syringe with needle, sterile 2 cc, each                                                                                                                                                                                                               |
| A4208                                                     | Syringe with needle, sterile 3 cc, each                                                                                                                                                                                                               |
| A4209                                                     | Syringe with needle, sterile 5 cc or greater, each                                                                                                                                                                                                    |
| A4211                                                     | Supplies for self-administered injections                                                                                                                                                                                                             |
| A4212                                                     | Non-coring needle or stylet with or without catheter                                                                                                                                                                                                  |
| A4213                                                     | Syringe, sterile, 20 cc or greater, each                                                                                                                                                                                                              |
| A4215                                                     | Needle, sterile, any size, each                                                                                                                                                                                                                       |
| A4221                                                     | Supplies for maintenance of drug infusion catheter, per week (list drug separately)                                                                                                                                                                   |

| Code  | Code Description                                                                                                                            |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------|
| A4222 | Infusion supplies for external drug infusion pump, per cassette or bag (list drugs separately)                                              |
| A4230 | Infusion set for external insulin pump, non-needle cannula type                                                                             |
| A4231 | Infusion set for external insulin pump, needle type                                                                                         |
| A4232 | Syringe with needle for external insulin pump, sterile, 3cc                                                                                 |
| A4233 | Replacement battery, alkaline (other than J cell), for use with medically necessary home blood glucose monitor owned by patient, each       |
| A4234 | Replacement battery, alkaline, J cell, for use with medically necessary home blood glucose monitor owned by patient, each                   |
| A4235 | Replacement battery, lithium, for use with medically necessary home blood glucose monitor owned by patient, each                            |
| A4236 | Replacement battery, silver oxide, for use with medically necessary home blood glucose monitor owned by patient, each                       |
| A4238 | Supply allowance for adjunctive continuous glucose monitor (cgm), includes all supplies and accessories, 1 month supply = 1 unit of service |
| A4244 | Alcohol or peroxide, per pint                                                                                                               |
| A4245 | Alcohol wipes, per box                                                                                                                      |
| A4246 | Betadine or pHisoHex solution, per pint                                                                                                     |
| A4247 | Betadine or iodine swabs/wipes , per box                                                                                                    |
| A4250 | Urine test or reagent strips or tablets (100 tablets or strips)                                                                             |
| A4252 | Blood ketone test or reagent strip, each                                                                                                    |
| A4253 | Blood glucose test or reagent strips for home blood glucose ,monitor, per 50 strips                                                         |
| A4255 | Platforms for home blood glucose monitor, 50 per box                                                                                        |

| Code  | Code Description                                                                                     |
|-------|------------------------------------------------------------------------------------------------------|
| A4256 | Normal, low, and high calibrator solution/chips                                                      |
| A4258 | Spring-powered device for lancet, each                                                               |
| A4259 | Lancets, per box of 100                                                                              |
| A4271 | Integrated lancing and blood sample testing cartridges for home blood glucose monitor, per month     |
| A9274 | External ambulatory insulin delivery system, disposable, each, includes all supplies and accessories |
| A9275 | Home glucose disposable monitor, includes test strips                                                |
| E0607 | Home blood glucose monitor                                                                           |
| E0784 | External ambulatory infusion pump, insulin                                                           |
| E2100 | Blood glucose monitor with integrated voice synthesizer                                              |
| E2101 | Blood glucose monitor with integrated lancing/blood sample                                           |
| E2102 | Adjunctive continuous glucose monitor or receiver                                                    |
| E2104 | Home blood glucose monitor for use with integrated lancing/blood sample testing cartridge            |
| G0108 | Diabetes outpatient self-management training services, individual, per 30 minutes                    |
| G0109 | Diabetes outpatient self-management training services, group session (2 or more), per 30 minutes     |
| J1811 | Insulin (fiasp) for administration through dme (i.e., insulin pump) per 50 units                     |
| J1812 | Insulin (fiasp), per 5 units                                                                         |
| J1813 | Insulin (lyumjev) for administration through dme (i.e., insulin pump) per 50 units                   |
| J1814 | Insulin (lyumjev), per 5 units                                                                       |

| Code  | Code Description                                                                                                                                                                                                                       |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| J1815 | Injection, insulin, per 5 units                                                                                                                                                                                                        |
| J1817 | Insulin for administration through DME (i.e., insulin pump) per 50 units                                                                                                                                                               |
| S5550 | Insulin, rapid onset, 5 units                                                                                                                                                                                                          |
| S5551 | Insulin, most rapid onset (Lispro or Aspart); 5 units                                                                                                                                                                                  |
| S5552 | Insulin, intermediate acting (NPH or LENTE); 5 units                                                                                                                                                                                   |
| S5553 | Insulin, long acting; 5 units                                                                                                                                                                                                          |
| S5560 | Insulin delivery device, reusable pen; 1.5 ml size                                                                                                                                                                                     |
| S5561 | Insulin delivery device, reusable pen; 3 ml size                                                                                                                                                                                       |
| S5565 | Insulin cartridge for use in insulin delivery device other than pump; 150 units                                                                                                                                                        |
| S5566 | Insulin cartridge for use in insulin delivery device other than pump; 300 units                                                                                                                                                        |
| S5570 | Insulin delivery device, disposable pen (including insulin); 1.5 ml size                                                                                                                                                               |
| S5571 | Insulin delivery device, disposable pen (including insulin); 3 ml size                                                                                                                                                                 |
| S8490 | Insulin syringes (100 syringes, any size)                                                                                                                                                                                              |
| S9140 | Diabetic management program, follow-up visit to non-MD provider                                                                                                                                                                        |
| S9141 | Diabetic management program, follow-up visit to MD provider                                                                                                                                                                            |
| S9145 | Insulin pump initiation, instruction in initial use of pump (pump not included)                                                                                                                                                        |
| S9353 | Home infusion therapy, continuous insulin infusion therapy; administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits coded separately), per diem |
| S9455 | Diabetic management program, group session                                                                                                                                                                                             |
| S9460 | Diabetic management program, nurse visit                                                                                                                                                                                               |
| S9465 | Diabetic management program, dietitian visit                                                                                                                                                                                           |

| Code                                                                                                                                                           | Code Description                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HCPCS codes not covered for indications listed in the CPB:                                                                                                     |                                                                                                                                                                                                                                                       |
| <i>Remote Glucose monitor, MiniMed Connect, Skin microvascular flow-motion</i> - no specific code [e.g. Dexcom SHARE - no additional reimbursement provided] : |                                                                                                                                                                                                                                                       |
| A4210                                                                                                                                                          | Needle-free injection device, each                                                                                                                                                                                                                    |
| A4257                                                                                                                                                          | Replacement lens shield cartridge for use with laser skin piercing device, each                                                                                                                                                                       |
| A9280                                                                                                                                                          | Alert or alarm device, not otherwise classified [hypoglycemic wristband alarm (e.g., Sleep Sentry)]                                                                                                                                                   |
| C1788                                                                                                                                                          | Port, indwelling (implantable)                                                                                                                                                                                                                        |
| E0620                                                                                                                                                          | Skin piercing device for collection of capillary blood, laser, each                                                                                                                                                                                   |
| ICD-10 codes covered if selection criteria are met:                                                                                                            |                                                                                                                                                                                                                                                       |
| E08.00 - E13.9                                                                                                                                                 | Diabetes mellitus                                                                                                                                                                                                                                     |
| G25.82                                                                                                                                                         | Stiff-man syndrome [indicated for GAD antibodies]                                                                                                                                                                                                     |
| H54.0X33 - H54.8                                                                                                                                               | Blindness and low vision                                                                                                                                                                                                                              |
| O24.011 - O24.93                                                                                                                                               | Diabetes mellitus in pregnancy, childbirth, and the puerperium                                                                                                                                                                                        |
| O99.810 - O99.815                                                                                                                                              | Abnormal glucose complicating pregnancy, childbirth, and the puerperium                                                                                                                                                                               |
| <i>Continuous Glucose Monitoring Devices (e.g., Dexcom, Eversense, Freestyle Libre, Guardian):</i>                                                             |                                                                                                                                                                                                                                                       |
| <i>Short-term monitoring [72 hours to 1 week] :</i>                                                                                                            |                                                                                                                                                                                                                                                       |
| CPT codes covered if selection criteria are met:                                                                                                               |                                                                                                                                                                                                                                                       |
| 95249                                                                                                                                                          | Ambulatory continuous glucose monitoring of interstitial tissue fluid via a subcutaneous sensor for a minimum of 72 hours; patient-provided equipment, sensor placement, hook-up, calibration of monitor, patient training, and printout of recording |
| 95250                                                                                                                                                          | Ambulatory continuous glucose monitoring of interstitial tissue fluid via a subcutaneous sensor for a minimum of 72 hours; sensor placement, hook-up, calibration of monitor, patient training, removal of sensor, and printout of recording          |

| Code                                                           | Code Description                                                                                                                                                                                                                                      |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 95251                                                          | analysis, interpretation and report                                                                                                                                                                                                                   |
| ICD-10 codes covered if selection criteria are met:            |                                                                                                                                                                                                                                                       |
| E08.00 - E13.9                                                 | Diabetes mellitus                                                                                                                                                                                                                                     |
| <i>Long-term monitoring [greater than 1 week]:</i>             |                                                                                                                                                                                                                                                       |
| CPT codes covered if selection criteria are met :              |                                                                                                                                                                                                                                                       |
| 0446T                                                          | Creation of subcutaneous pocket with insertion of implantable interstitial glucose sensor, including system activation and patient training                                                                                                           |
| 0447T                                                          | Removal of implantable interstitial glucose sensor from subcutaneous pocket via incision                                                                                                                                                              |
| 0448T                                                          | Removal of implantable interstitial glucose sensor with creation of subcutaneous pocket at different anatomic site and insertion of new implantable sensor, including system activation                                                               |
| 95249                                                          | Ambulatory continuous glucose monitoring of interstitial tissue fluid via a subcutaneous sensor for a minimum of 72 hours; patient-provided equipment, sensor placement, hook-up, calibration of monitor, patient training, and printout of recording |
| 95250                                                          | Ambulatory continuous glucose monitoring of interstitial tissue fluid via a subcutaneous sensor for a minimum of 72 hours; sensor placement, hook-up, calibration of monitor, patient training, removal of sensor, and printout of recording          |
| 95251                                                          | analysis, interpretation and report                                                                                                                                                                                                                   |
| HCPCS codes covered if selection criteria are met:             |                                                                                                                                                                                                                                                       |
| <i>Eversense E3 implantable CGM sensor - no specific code:</i> |                                                                                                                                                                                                                                                       |
| A4239                                                          | Supply allowance for non-adjunctive, non-implanted continuous glucose monitor (cgm), includes all supplies and accessories, 1 month supply = 1 unit of service                                                                                        |



| Code                                                 | Code Description                                                                                                                                                                                |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A9276                                                | Sensor; invasive (e.g., subcutaneous), disposable, for use with interstitial continuous glucose monitoring system, 1 unit = 1 day supply                                                        |
| A9277                                                | Transmitter; external, for use with interstitial continuous glucose monitoring system                                                                                                           |
| A9278                                                | Receiver (monitor); external, for use with interstitial continuous glucose monitoring system                                                                                                    |
| E2103                                                | Non-adjunctive, non-implanted continuous glucose monitor or receiver                                                                                                                            |
| G0564                                                | Creation of subcutaneous pocket with insertion of 365 day implantable interstitial glucose sensor, including system activation and patient training                                             |
| G0565                                                | Removal of implantable interstitial glucose sensor with creation of subcutaneous pocket at different anatomic site and insertion of new 365 day implantable sensor, including system activation |
| S1030                                                | Continuous noninvasive glucose monitoring device, purchase                                                                                                                                      |
| S1031                                                | Continuous noninvasive glucose monitoring device, rental, including sensor, sensor replacement, and download to monitor                                                                         |
| ICD-10 codes covered if selection criteria are met : |                                                                                                                                                                                                 |
| E10.10 - E10.9                                       | Type 1 diabetes mellitus                                                                                                                                                                        |
| E11.00 - E11.9                                       | Type 2 diabetes mellitus [is considered experimental and investigational for persons not using intensive insulin regimens]                                                                      |
| E74.00                                               | Glycogen storage disease, unspecified                                                                                                                                                           |
| E74.01                                               | Von Gierke disease [type 1a glycogen storage disease]                                                                                                                                           |
| E74.02                                               | Pompe disease                                                                                                                                                                                   |
| E74.03                                               | Cori disease                                                                                                                                                                                    |
| E74.04                                               | McArdle disease                                                                                                                                                                                 |

| Code                                                        | Code Description                                                          |
|-------------------------------------------------------------|---------------------------------------------------------------------------|
| E74.05                                                      | Lysosome-associated membrane protein 2 [LAMP2] deficiency                 |
| E74.09                                                      | Other glycogen storage disease                                            |
| ICD-10 codes not covered for indications listed in the CPB: |                                                                           |
| E16.9                                                       | Disorder of pancreatic internal secretion, unspecified [nesidioblastosis] |
| O24.410 - O24.439                                           | Gestational diabetes mellitus                                             |
| P70.3 - P70.4                                               | Iatrogenic and other neonatal hypoglycemia                                |

## Background

### Diabetes Test Strips and Meters

Diabetic test strips and meters are test strips and meters used for blood sugar testing. Once a blood sample has made it on to the glucose strip, a glucose meter device is used to measure the glucose in the blood. In each test strip, glucose oxidase reacts with the glucose in the blood sample to form gluconic acid. Diabetic test strips and meters are available in multiple brands. Examples of brand names include: Abbott Optium Plus; Abbott Freestyle; Accu-Chek Aviva; Accu-Chek Active; Accu-Chek Advantage; Accu-Chek Compact; Bayer Contour; Bayer Contour NEXT; OneTouch Ultra; OneTouch VerioFlex; and OneTouch Verio.

## Glutamic Acid Decarboxylase (GAD-65) Antibodies

Glutamic acid decarboxylase (GAD) is an enzyme produced primarily by pancreatic islet cells. A number of recent studies indicate that patients with type 1 diabetes often have antibodies to GAD and several other islet cell antigens. This is consistent with the hypothesis that type 1 diabetes is an autoimmune disease and that autoantibody production is an early step in the development of type 1 diabetes. Autoantibodies can be detected in many cases prior to the onset of glucose intolerance. The presence of GAD autoantibodies has been shown to be a strong predictive marker for the eventual onset of type 1 diabetes.

Measurement of anti-GAD antibodies has been proposed for evaluating the risk of developing type 1 diabetes in persons at high risk. However, the value of such testing is unproven, as there are no measures that have been demonstrated to be effective in preventing the onset of type 1 diabetes. Guidelines from the Canadian Diabetes Association (Ur et al., 2003) explain that the loss of pancreatic beta cells in persons who subsequently develop type 1 diabetes passes through a subclinical prodrome that can be detected reliably in first- and second-degree relatives of persons with type 1 diabetes by the presence of anti-GAD antibodies and other pancreatic islet cell autoantibodies in their sera. While randomized trials testing prevention strategies have been completed or are now underway, safe and effective preventive strategies have not been identified. The guidelines concluded: "Therefore, any attempts to prevent type 1 diabetes should be undertaken only within the confines of formal research protocols."

Measurement of anti-GAD antibody can be useful in distinguishing type 1 from type 2 diabetes when the clinical history is ambiguous. Guidelines from the Royal Australian College of General Practitioners (RACGP, 2007) explained that measurement of GAD can be particularly useful in diagnosing Late Onset Autoimmune Diabetes in Adults (LADA), a form of late-onset diabetes that is autoimmune and requires treatment with insulin within a relatively short period after diagnosis (often within the next 2 years). RACGP guidelines explained that persons with

LADA tend to be young (30 to 40 years of age), lean, and have a personal and/or family history of other autoimmune diseases (e.g., hypo- or hyperthyroidism). The guidelines stated that testing for GAD antibodies can confirm the diagnosis in ambiguous cases and prompt counseling about the likely time course of diabetes progression and the possibility of other autoimmune diseases. In addition, the establishment of the LADA diagnosis may be useful in selecting therapy (Brophy et al., 2007).

Antibodies to GAD are often markedly elevated in patients with stiff-person syndrome (also referred to as stiff-man syndrome), a condition associated with fluctuating stiffness and paroxysmal spasms of the trunk and legs.

### Jet Injectors

Jet injectors offer an alternative method of hypodermic drug delivery compared to conventional needle-syringe systems. The main objective of using a jet injector, as opposed to a conventional needle-syringe, is to increase patient comfort during injection. A review of the literature indicates that some patients prefer injection by jet injector, while others may experience more discomfort. The Youth Task Force of the American Diabetes Association (Task Force on Jet Injections, 1998, 1991) reviewed the scientific literature on jet injection and could not make general recommendations for their use due to insufficient information. There are hypothetical risks and benefits associated with their use for insulin delivery that have not been clearly addressed in the literature.

There is insufficient information on the frequency and significance of aversion to needles; therefore, the use of jet injectors due to a fear of needles is a matter of convenience and patient preference. A jet injector may be appropriate for some individuals with medical conditions that make it impossible for them to use a conventional needle-syringe. The specific

group of patients unable to use syringes and who are therefore candidates for jet injections based on medical necessity has not been defined in the literature, but it reasonably includes individuals with severe arthritis, severe tremors, or blindness.

### Continuous Glucose Monitoring (CGM) Devices

Continuous glucose monitoring systems (CGMS) are devices that measure glucose levels in interstitial fluid at programmable intervals. These readings, used along with fingerstick results, help detect any patterns or trends in an individual's glucose levels and are intended to assist in calculating the insulin dosage needed to manage glycemic control. CGMS readings are intended to supplement, not replace, finger sticks.

CGMS use sensors that are inserted under the skin in the abdomen and work by extracting glucose from the interstitial fluid, measuring and recording the glucose level, and converting these measurements into equivalent blood glucose readings.

With certain CGMS (sometimes referred to as "professional CGM"), the patient receives no information while wearing the device. Results can be determined in a clinician's office and graphed to provide information about the frequency of unrecognized hypoglycemia and the extent of within-day and between-day variations in blood glucose.

Other devices (often referred to as "personal CGM") provide the patient with real-time results of glucose values on a continuous basis, which can be automatically and securely shared with a clinician via a mobile medical glucose-monitoring application downloaded onto a mobile device, such as a cell phone.

Many sensors are designed to be worn for three to seven days, depending on the product. Calibration is required whenever a new glucose sensor is inserted, which necessitates obtaining blood glucose from a traditional fingerstick sample.

The MiniMed Continuous Subcutaneous Glucose Monitoring System is an example of a diagnostic continuous glucose monitor for short-term use. The FDA granted the MiniMed CGMS (Medtronic MiniMed, Minneapolis, MN) pre-market approval in June 1999 for use as an adjunct to finger-stick blood glucose testing. CGMS uses a recorder and subcutaneously implanted glucose sensor to store data to be downloaded to a personal computer. The system provides continuous measurements of the interstitial glucose levels that range from 40 to 400 mg/dL. The glucose sensor signal is acquired every ten seconds. An average of the acquired signals is saved in memory every five minutes for up to three days. While in operation, the MiniMed CGMS monitor does not display glucose values, and individuals are still required to test their glucose levels several times a day by a standard method (finger sticks) and enter the glucose measurements into the monitor for calibration purposes. The system is used as a diagnostic tool to evaluate glucose levels over a three-day period and is then returned to the physician for evaluation of results to potentially modify diabetes treatment regimens; it is not intended for long-term use. According to the FDA, the MiniMed CGMS is not intended to replace standard finger-stick testing. The newest short-term CGMS version of the MiniMed is the iPRO2.

More recently, the FDA approved the Guardian Real-Time (RT) Continuous Glucose Monitoring System (Medtronic, Minneapolis, MN), which is described by the manufacturer as the first consumer continuous glucose monitoring device. The Guardian CGMS device uses a glucose sensor connected to a transmitter that sends glucose readings every five minutes to a monitor. The glucose sensor is typically discarded and replaced after three days. According to the manufacturer, the device provides up to 288 glucose readings per day or every five minutes. Unique features include predictive and rate-of-change alarms and expanded trend graphs. Graphs can show the effect of exercise, diet, and lifestyle, as well as medication on glucose values using 3, 6, 12, and 24-hour increments. Alarms that signal high and low glucose alerts warn individuals of any significant glucose changes. Data is downloaded using the Medtronic Carelink Therapy Management Software. According to the FDA-approved

labeling, the Guardian RT is indicated to supplement blood glucose information from standard home blood glucose meters for persons 7 years and older with type 1 or type 2 diabetes. A fingerstick measurement is required before taking action.

The DexCom STS Continuous Glucose Monitoring System (DexCom, Inc., San Diego, CA) gained FDA approval on March 24, 2006. It is a glucose sensor that reports glucose values every five minutes for up to 72 hours. These readings are used with fingerstick results to detect trends and patterns in glucose levels in adults with diabetes aged 18 years and over. The DexCom STS is indicated for use as an adjunctive device to complement, not replace, information obtained from standard home glucose monitoring devices.

The Dexcom G4 PLATINUM (adult and pediatric) Systems with Share is a CGMS that incorporates a sensor, transmitter, and receiver, allowing glucose to be monitored every five minutes. The sensor reports glucose for seven days before a new sensor replacement is necessary. A built-in alarm system can be programmed by the user when glucose falls below a pre-set low and a pre-set high level. A built-in hypoglycemia safety alarm alerts the user when glucose hits 55 mg/dL. It is FDA approved for use in adults and children ages 2 years and older. Dexcom Share is an integrated wireless communication system that is built into the receiver, enabling remote monitoring capabilities and sharing of data through a compatible internet-accessible device application.

The Dexcom G5 Mobile CGM System is described as a complete, mobile CGMS with wireless technology that is built into the transmitter. Glucose information is sent wirelessly from the device transmitter to a compatible smart device or Dexcom G5 receiver. It is FDA approved for use in adults and children ages 2 years and older.

In 2018, the US Food and Drug Administration (FDA) approved the Dexcom G6 device to replace the Dexcom G5 system. The Dexcom G6 system can be integrated with an automated insulin dosing system, such that a rise in blood glucose would trigger the release of insulin

from an insulin pump without the need for confirmatory fingerstick glucose testing. Fingerstick glucose determinations are also not needed for calibration. In a study using an older version of the Dexcom device, the use of CGM without confirmatory self-monitoring of blood glucose (SMBG) was as safe and effective as using CGM with SMBG. Older Dexcom CGMs and the Enlite CGM do not have a non-adjunctive claim, so fingerstick glucose readings are required both for calibration and as confirmation of glucose levels before making decisions on insulin dosing.

Combined external insulin pumps with CGMS are devices that integrate an insulin pump with real-time continuous glucose monitoring and are not intended to replace finger sticks. These systems incorporate features including predictive alerts that give early warnings so action can be taken to prevent dangerous high or low blood glucose events.

The Paradigm Real Time System (Medtronic MiniMed) is an open-loop insulin delivery system that combines an external insulin pump with continuous monitoring of interstitial glucose levels via a subcutaneous sensor. The sensor communicates glucose readings to the pump using a radio transmitter. The pump can also calculate recommended insulin doses, which the patient can accept or modify. Readings from the continuous glucose monitor (CGM) are not intended to be used to make therapy adjustments. A conventional blood glucose meter reading is needed before making adjustments because there is a lag of up to 10 minutes in glucose concentration in the interstitial fluid relative to the concentration in the blood (CADTH, 2007). Furthermore, readings from the sensor may be less accurate in the hypoglycemic range. An assessment of the Paradigm Real Time System by the Canadian Agency for Drugs and Technologies in Health (CADTH, 2007) concluded: "Based on the limited amount of research published to date, the impact of the Paradigm Real-Time System on long-term glycemic control, prevention of diabetic complications, or quality of life is unclear." The assessment noted that open-loop systems such as the Paradigm Real Time System are an incremental step towards a fully closed-loop system, also known as an artificial pancreas, where insulin dosages would be automatically adjusted rather than requiring patient input.



The MiniMed Paradigm REAL-Time Revel System is approved for use in adults and children ages 7 years and older and works with the mySentry remote monitoring system. The mySentry remote monitoring system is an optional monitoring device for the MiniMed Paradigm REAL-Time Revel System. The mySentry consists of a remote outpost and monitor. Blood glucose levels collected by the CGM are sent to the remote (wireless) monitor, which also has alarms to alert the user of high or low blood glucose levels.

The Animas Vibe System combines a Dexcom G4 Platinum sensor and transmitter with the Animas Vibe insulin pump. It is FDA approved for use in adults and children ages 2 years and older.

Guidelines from the National Institute for Health and Clinical Excellence (2004) recommended the use of CGM devices for the evaluation of persons with type 1 diabetes on insulin therapy who have repeated hypoglycemia and hyperglycemia at the same time each day, and hypoglycemia unawareness unresponsive to conventional insulin dose adjustment.

A Cochrane systematic evidence review found limited evidence for the effectiveness of real-time continuous glucose monitoring (CGM) use in children, adults, and patients with poorly controlled diabetes (Miranda et al., 2012). The authors found that the risk of severe hypoglycemia or ketoacidosis was not significantly increased for CGM users, but as these events occurred infrequently, these results have to be interpreted cautiously.

A multi-center randomized clinical study sponsored by the Juvenile Diabetes Research Foundation (JDRF, 2008) provided evidence of improved glycemic control over the intermediate term (6 months) with the use of CGMs in adults greater than 24 years of age. In this study, 322 adults and children who were already receiving intensive therapy for type 1 diabetes were randomly assigned to a group with continuous glucose monitoring or to a control group performing home monitoring with a blood glucose meter. All subjects were stratified into three groups according to age and had a glycated hemoglobin level of 7.0 to

10.0%. The investigators found that the changes in glycated hemoglobin levels in the two study groups varied markedly according to age group, with a significant benefit in patients 25 years of age or older with continuous glucose monitoring (mean difference in change,  $-0.53\%$ ,  $p < 0.001$ ). There was no significant benefit in glycated hemoglobin levels with continuous glucose monitoring in subjects who were 15 to 24 years of age (mean difference,  $0.08$ ;  $p = 0.52$ ) or among those who were 8 to 14 years of age (mean difference,  $-0.13$ ;  $p = 0.29$ ). The investigators posited that the disparate outcomes may be due to poorer compliance among younger age groups. The use of continuous glucose monitoring averaged 6.0 or more days per week for 83% of patients 25 years of age or older, 30% of those 15 to 24 years of age, and 50% of those 8 to 14 years of age. The study also found no significant difference in the rate of severe hypoglycemia among persons who were assigned to CGMs and those who performed home monitoring with a blood glucose meter; however, the investigators noted that the trial was not powered to detect such a difference. Commenting on the JDRF study of continuous glucose monitoring, Brett (2008) noted that "although this method is appealing theoretically, the extent to which it will improve long-term clinical outcomes remains to be determined."

A controlled clinical study, the Sensor-Augmented Pump Therapy for A1C Reduction (STAR) 3 trial, found that, in both adults and children with inadequately controlled type 1 diabetes, sensor-augmented pump therapy resulted in significant improvement in glycated hemoglobin levels compared with injection therapy (Bergenstal et al., 2010). However, whether these results can be translated into community practice outside of the controlled clinical trial setting is unknown. A 1-year, multi-center, randomized, controlled trial compared the efficacy of sensor-augmented pump therapy (pump therapy) with that of a regimen of multiple daily insulin injections (injection therapy) in 485 patients (329 adults and 156 children) with inadequately controlled type 1 diabetes. Patients received recombinant insulin analogs and were supervised by expert clinical teams. The primary endpoint was the change from the baseline glycated hemoglobin level. At 1 year, the baseline mean glycated hemoglobin level ( $8.3\%$  in the two study groups) had decreased to  $7.5\%$  in the pump-therapy group, compared with  $8.1\%$  in the injection-therapy group ( $p < 0.001$ ). The proportion of patients who reached

the glycated hemoglobin target (less than 7%) was greater in the pump-therapy group than in the injection-therapy group. The rate of severe hypoglycemia in the pump-therapy group (13.31 cases per 100 person-years) did not differ significantly from that in the injection-therapy group (13.48 per 100 person-years,  $p = 0.58$ ). There was no significant weight gain in either group. A commentator noted that this is a manufacturer-sponsored study, and the investigators included employees of the firm. An editorialist (Wolpert, 2010) offered several caveats: These patients received intensive support and monitoring that are not available to many patients, were highly skilled at self-management before enrollment, and had to be adept at calibration and management of equipment failures and alarms. The editorialist warned that the "expert training and guidance [on the use of continuous glucose monitoring] received by patients in clinical trials cannot be readily duplicated in a busy clinical practice." A commentary (Schwenk, 2010) concluded: "This new and expensive technology must be tested in wider community-based trials before it will be ready for broad dissemination."

The editorialist (Wolpert, 2010) also compared the results of the STAR-3 trial with the JDRF trial and stated that the differences in outcome may be due to differences in the design of these trials. In the STAR 3 study, the patients in the pump-therapy group changed their mode of both insulin delivery and glucose monitoring at the time of randomization, whereas in the JDRF trial, patients who were assigned to receive continuous glucose monitoring did not change their mode of insulin delivery. The editorialist stated that the greater reduction in glycated hemoglobin levels among adult patients in the STAR 3 trial than in the JDRF study may reflect the additional effect of initiating pump therapy, as well as the increased baseline glycated hemoglobin levels in the STAR 3 study compared with the JDRF study (8.3% and 7.6%, respectively).

The editorialist (Wolpert, 2010) also noted that, in the STAR 3 study, the improved glycemic control among children in the pump-therapy group contrasted with the lack of apparent benefit for continuous glucose monitoring among children in the JDRF trial. The editorialist questioned whether the benefits seen in the pump-therapy group in the STAR 3 study were due primarily

to the initiation of pump therapy rather than to continuous glucose monitoring. The editorialist explained that consistent with this possibility is the fact that patients in the STAR 3 study who had a relatively low frequency of sensor use had significant improvements in glycated hemoglobin levels. The editorialist noted that children in the injection-therapy group, who used intermittent capillary blood glucose monitoring, had lower rates of both severe and biochemical hypoglycemia than did patients in the JDRF trial who used continuous monitoring. The editorialist said that these results suggest that the selection of patients may also account for some of the differences in the outcomes of these two trials.

Whether the benefits of CGM in improving glycemic control extend beyond the intermediate term (12 months) is unknown. A large clinical study of CGM, the Minimally Invasive Technology Role and Evaluation (MITRE) study, sponsored by the National Institute for Health Research Health Technology Assessment Program, found that continuous blood glucose monitoring had no durable effect on blood glucose control (Newman et al., 2007; Newman et al., 2009). The purpose of the MITRE study was to evaluate the efficacy of minimally invasive glucose monitoring devices in 400 patients with diabetes mellitus treated with insulin. The primary endpoint was long-term glucose control, as indicated by changes in glycosylated hemoglobin (HbA1c) levels for 18 months. A total of 400 patients were randomly assigned to the CGMS by MiniMed, the Biographer by Animas, a standard control, or to an attention control group. Mean baseline HbA1c ranged from 7.0% to 15.5% for participants. All groups demonstrated a decline in mean HbA1c, especially during the first few months of the study. However, by month 18, the percentage of patients that had a relative reduction of at least 12.5% was 15% in the Biographer group, 27% in the CGMS group, 24% in the standard control, and 27% in the attention control group. The relative decline in HbA1c from baseline ranged from 1% to 4.6%. The results suggested that the use of the CGMS had a small benefit, but only in the short term, and that the Biographer had less impact on HbA1c than either the CGMS or standard treatment. The assessment concluded: "Continuous glucose monitors as assessed in this study do not lead to improved clinical outcomes and are not cost-effective for improving HbA1c in unselected individuals with poorly controlled insulin-requiring diabetes"

(Newman et al., 2009). Some commentators have posited that more advanced continuous glucose monitoring devices currently in use may provide more durable results than the monitors used in the MITRE study. The results from CGMs used in this study were downloaded and reviewed with the endocrinologist, but only the Glucowatch Biographer provided a real-time display of glucose results to the patient (the MiniMed CGMS used in this study did not include a real-time display of glucose readings). Whether the more advanced CGMs with real-time display will provide more durable results than earlier models used in the MITRE study is a question for future long-term studies.

Norgaard et al. (2013) reported on the largest and longest multicenter prospective observational study of continuous glucose monitoring with insulin infusion pumps, so-called sensor-augmented pump therapy. The investigators reported on a 12-month observational study in patients with type 1 diabetes treated with continuous subcutaneous insulin infusion (CSII) upon the introduction of continuous glucose monitoring (CGM). The study was conducted in 15 countries to document the real-life use of sensor-augmented pump therapy and assess which variables are associated with improvement in type 1 diabetes management. Data from 263 patients (38% male; mean age,  $28.0 \pm 15.7$  years [range, 1-69 years]; body mass index,  $23.3 \pm 4.9$  kg/m<sup>2</sup>; diabetes duration,  $13.9 \pm 10.7$  years; CSII duration,  $2.6 \pm 3$  years) were collected. Baseline mean glycated hemoglobin A1c (HbA1c) was  $8.1 \pm 1.4\%$ ; 82% had suboptimal HbA1c ( $\geq 7\%$ ). The investigators found that the average sensor use for 12 months was only 30% (range, 0-94%), and that sensor use decreased with time (first 3 months, 37%; last 3 months, 27%). The investigators found that there were significantly more patients with an HbA1c value of  $< 7.5\%$  after 3 months of sensor-augmented pump therapy than at baseline (baseline, 29%; 3 months, 37%). However, the percentage of patients with an HbA1c value of  $< 7.5\%$  decreased over the 12-month observation period, such that the percentage of patients with an HbA1c value of  $< 7.5\%$  after 12 months was not statistically significantly higher than at baseline.

A published systematic evidence review and meta-analysis of the evidence for continuous glucose monitoring systems in children with type 1 diabetes reached the following conclusions (Golicki et al., 2008): "The Continuous Glucose Monitoring System is not better than self-monitoring of blood glucose with regard to improvement of metabolic control among type 1 diabetic children. However, due to the small number of participants and methodological limitations of the studies included, findings of this meta-analysis should be interpreted with caution."

Chetty et al. (2008) of McMaster University performed a meta-analysis of randomized controlled trials comparing continuous glucose monitoring and self-blood fingerstick glucose monitoring in persons with type 1 diabetes. The investigators found insufficient evidence to support the notion that CGM provides a superior benefit over self-blood fingerstick glucose monitoring in terms of hemoglobin A1c reduction. The investigators, however, found some indication of improved detection of asymptomatic nocturnal hypoglycemia in the CGM group. The investigators identified seven studies with a total of 335 patients that fulfilled the inclusion criteria. Five studies were confined to the pediatric population (age less than 18 years). Study duration varied from 12 to 24 weeks. The investigators found that, compared with self-blood fingerstick glucose monitoring, CGM was associated with a non-significant reduction in hemoglobin A1c (0.22%; 95% confidence interval [CI]: -0.439% to 0.004%,  $p = 0.055$ ).

Regarding the therapeutic use of continuous glucose monitoring devices for hypoglycemic unawareness, current evidence from randomized controlled clinical trials has focused on CGM's effect on shortening the duration of asymptomatic hypoglycemia, an intermediate endpoint, rather than clinical outcomes. The clinical significance of reductions in the duration of asymptomatic hypoglycemia is unknown. In addition, current evidence indicates that continuous glucose monitoring devices are least accurate in the hypoglycemic range (CADTH, 2007; Melki et al., 2006).

Hypoglycemia unawareness is reversible. Meticulous avoidance of hypoglycemia for several weeks is sufficient to restore awareness of hypoglycemia (Cheng et al., 2000; Fanelli et al., 1993; Dagogo-Jack et al., 1994; Cranston et al., 1994). The return of awareness is accomplished with minimal compromise of glycemic control, but it requires substantial involvement of health professionals. In addition, unlike CGM, HAATT/BGAT (Hypoglycemia Anticipation, Awareness and Treatment Training/Blood Glucose Awareness Training) has been proven to reduce the occurrence of severe hypoglycemia (Cox et al., 2001; Cox et al., 2004).

There is limited evidence of the effectiveness of CGMs to improve outcomes in pregnant women with diabetes. Murphy et al. (2008) reported on an open-label randomized controlled clinical trial where 71 pregnant women with type 1 diabetes (n = 46) or type 2 diabetes (n = 25) were randomly assigned to antenatal care plus continuous glucose monitoring (n = 38) or to standard antenatal care (n = 33). Continuous glucose monitoring was used as an educational tool to inform shared decision-making and future therapeutic changes at intervals of 4 to 6 weeks during pregnancy. All other aspects of antenatal care were equal between the groups. Women randomized to continuous glucose monitoring had lower mean hemoglobin A1c levels (5.8%) from 32 to 36 weeks' gestation compared with women randomized to standard antenatal care (6.4%). Compared with infants of mothers in the control arm, those of mothers in the intervention arm had decreased mean birthweight standard deviation scores (0.9 versus 1.6), decreased median customized birthweight centiles (69% versus 93%), and a reduced risk of macrosomia (odds ratio 0.36). The investigators noted a number of limitations to this study. Although efforts were made to standardize antenatal contacts between groups, health professionals were not blinded, and therefore the possibility of bias in clinical management cannot be excluded. Differences in maternal characteristics, with longer duration of diabetes in the intervention group, may have contributed to some of the effect on infant outcomes. The investigators stated that the study included a small number of women and that larger multicenter trials are required to assess the impacts of continuous glucose monitoring in pregnancy.

Guidelines from the American Diabetes Association (2018) state that, when used properly, continuous glucose monitoring (CGM) in conjunction with intensive insulin regimens is a useful tool to lower A1C in adults with type 1 diabetes who are not meeting glycemic targets (A - recommendation based upon evidence from well-conducted, generalizable, randomized controlled trials that are adequately powered). The guidelines state that CGM may be a useful tool in those with hypoglycemia unawareness and/or frequent hypoglycemic episodes (C - recommendation based upon evidence from poorly controlled or uncontrolled studies). Given the variable adherence to CGM, assess individual readiness for continuing CGM use prior to prescribing (E - recommendation based upon expert consensus or clinical experience). When prescribing CGM, robust diabetes education, training, and support are required for optimal CGM implementation and ongoing use (E - based upon expert consensus or clinical experience). People who have been successfully using CGM should have continued access after they turn 65 years of age (E - based upon expert consensus or clinical experience).

A joint statement from the European Association for the Study of Diabetes and the American Diabetes Association Diabetes Technology Working Group stated that the first systems for continuous glucose monitoring (CGM) became available over 15 years ago. Many then believed CGM would revolutionize the use of intensive insulin therapy in diabetes; however, progress toward that vision has been gradual. Although increasing, the proportion of individuals using CGM rather than conventional systems for self-monitoring of blood glucose on a daily basis is still low in most parts of the world. Barriers to uptake include cost, measurement reliability (particularly with earlier-generation systems), human factors issues, lack of a standardized format for displaying results, and uncertainty on how best to use CGM data to make therapeutic decisions. This Scientific Statement makes recommendations for systemic improvements in clinical use and regulatory (pre- and post-marketing) handling of CGM devices. The aim is to improve safety and efficacy in order to support the advancement of the technology in achieving its potential to improve quality of life and health outcomes for more people with diabetes.



A structured review of the evidence conducted by the BlueCross BlueShield Association Technology Evaluation Center (2003) concluded that "use of intermittent or continuous interstitial fluid glucose monitoring in patients with diabetes mellitus does not meet Blue Cross and Blue Shield Association Technology Evaluation Center criteria." Similarly, a technology assessment conducted by the California Technology Assessment Forum (CTAF) concluded that continuous glucose monitoring does not meet CTAF's criteria (Tice, 2003). An updated assessment of continuous glucose monitoring by the California Technology Assessment Forum (Karliner, 2009) concluded that continuous glucose monitoring devices meet CTAF criteria for use in type 1 diabetes mellitus in non-pregnant adults requiring multiple (greater than or equal to 3) daily insulin injections and frequent (greater than or equal to 3) self-monitoring blood glucose checks. Continuous glucose monitoring devices did not meet CTAF criteria for the management of type 1 diabetes mellitus in children, adolescents, and pregnant women. The CTAF assessment explained that the largest randomized controlled clinical trial to date of continuous glucose monitoring devices for adults and children (citing JDRF, 2008) found conclusive benefit only for adults 25 years and older. The CTAF assessment explained that, while in this study, and in other smaller randomized controlled trials, there is evidence that both children and adults spend less time in a hypoglycemic glucose range when using a continuous glucose monitoring device compared to usual care frequent SMBG, there is little evidence that the use of a continuous glucose monitoring device confers an ultimate health benefit as measured by HbA1C as a marker of overall glycemic control. The CTAF assessment stated that it may be that for children and adolescents this is in large part due to difficulty with device adherence and not with the device itself. The CTAF assessment explained, however, that a health technology is only as good as its actual clinical application, and the evidence has not yet shown conclusive benefit for children, adolescents, and even young adults. Likewise, while the small studies that exist of pregnant women show the feasibility of continuous glucose monitoring device use during pregnancy, they do not yet demonstrate conclusive benefit in this population either. The CTAF assessment concluded that future study of these devices should incorporate more research on how the devices can be made more acceptable and user-friendly for children and adolescents with type 1 diabetes in

order to optimize potential clinical benefit for this population. The CTAF assessment also stated that larger studies of pregnant women, which are limited to those women requiring multiple insulin injections per day, are needed in order to adequately assess potential benefit in this population.

There are fewer trials evaluating CGM in patients with type 2 diabetes. In one meta-analysis, there was a significant reduction in A1C with the use of CGM versus SMBG in adults with type 2 diabetes (WMD -0.7 percent). In a subsequent trial, 158 adults treated with multiple daily injections of insulin (mean A1C 8.5 percent) were randomly assigned to CGM or usual care (SMBG at least four times daily). After 24 weeks, the reduction in A1C was greater with CGM (0.8 versus 0.5 percentage points, respectively; adjusted mean difference -0.3, 95% CI -0.5 to 0.0 percent). Patients in the CGM group performed a mean of 2.9 fingersticks daily, compared with 3.8 per day in the control group. There was no difference in hypoglycemia, which was infrequent in both groups, or in quality-of-life measures.

A technology assessment of self-monitoring of blood glucose in persons with type 2 diabetes prepared for the Centers for Medicare and Medicaid Services (Balk et al., 2006) commented that "currently, CGM [continuous glucose monitoring] has been studied primarily in children with type 1 diabetes. It is unclear whether CGM provides added value to traditional SMBG [self-monitoring of blood glucose]."

Available evidence shows that, in contrast to type 1 diabetes, persons with type 2 diabetes do not benefit from tight glucose control. In three recent large randomized trials (ACCORD, ADVANCE, and VADT), tight control in patients with long-standing type 2 diabetes did not lower overall mortality, cardiovascular-related mortality, stroke, amputations, or even clinical (as opposed to surrogate) microvascular endpoints (Gerstein et al., 2008; Patel et al., 2008; Duckworth et al., 2009). Some authorities suggested that the HbA1c goals for practice

guidelines should not be less than 7% and that, to encourage individualized treatment, performance measures should set an upper limit (e.g., 9%) rather than a lower limit (e.g., less than 7%) (Lehman and Krumholz, 2009).

A Cochrane systematic evidence review found that intensive glucose control significantly prevents the development of clinical neuropathy in type 1 diabetes mellitus (Callaghan et al., 2012). However, in type 2 diabetes mellitus, the effect of intensive glucose control on the incidence of clinical neuropathy was not statistically significant. The systematic evidence review also found that intensive glucose control significantly increases the risk of severe hypoglycemic episodes, which needs to be taken into account when evaluating its risk/benefit ratio.

A systematic evidence review of continuous glucose monitoring by the Ontario Ministry of Health and Long-Term Care Medical Advisory Secretariat (MAS, 2011) found that there was moderate quality evidence that in diabetic individuals with an infusion pump, continuous blood glucose monitoring plus self-monitoring was not more effective in reducing glycosylated hemoglobin, hypoglycemic events, or severe hypoglycemic events than self-monitoring alone.

A systematic evidence review prepared for the Agency for Healthcare Research and Quality (Golden et al., 2012) reported that randomized controlled trials showed no difference in the effect of CSII and MDI on HbA1c (moderate strength of evidence [SOE]) or severe hypoglycemia (low SOE) for children or adolescents with type 1 diabetes, or for adults with type 2 diabetes. In adults with type 1 diabetes, HbA1c decreased more with CSII than with MDI (low SOE), but results were heavily influenced by one study. The assessment found that there was no difference in severe hypoglycemia (low SOE). In children and adults with type 1 diabetes, CSII use was associated with improved quality of life compared with MDI (low SOE). There was insufficient evidence about quality of life for adults with type 2 diabetes. The SOE regarding pregnant women with pre-existing diabetes was either low or insufficient on all outcomes.

A systematic evidence review (Coca et al., 2012) found that intensive glucose control reduced the risk for microalbuminuria and macroalbuminuria in persons with type 2 diabetes, but evidence was lacking that it reduced the risk of significant clinical renal outcomes, such as doubling of the serum creatinine level, end-stage renal disease, or death from renal disease during the years of follow-up of the trials.

A review of continuous glucose monitors in type 2 diabetes by Meade (2012) stated that only five of the studies reviewed documented a reduction in HbA1c, and of these five, only three focused exclusively on patients with type 2 diabetes. The review stated that the majority of the studies evaluating CGM use in patients with type 2 diabetes were not designed to show a reduction in HbA1c. Studies of patients with type 2 diabetes mellitus had a smaller sample size than the studies reviewing continuous glucose monitoring in patients with type 1 diabetes.

Evidence to support the use of continuous glucose monitoring in persons with diabetes not on insulin is very limited. Vigersky et al. (2011) reported on a randomized controlled trial of 100 adults with type 2 diabetes who were not on prandial insulin. This study compared the effects of 12 weeks of intermittent continuous glucose monitoring with self-monitoring of blood glucose (SMBG) on glycemic control over a 40-week follow-up period. Subjects received diabetes care from their regular provider without therapeutic intervention from the study team. The investigators reported that there was a significant difference in A1C at the end of the 3-month active intervention that was sustained during the follow-up period. The mean, unadjusted A1C decreased by 1.0, 1.2, 0.8, and 0.8% in the continuous glucose monitoring group versus 0.5, 0.5, 0.5, and 0.2% in the SMBG group at 12, 24, 38, and 52 weeks, respectively ( $p = 0.04$ ). There was a significantly greater decline in A1C over the course of the study for the continuous glucose monitoring group than for the SMBG group, after adjusting for covariates ( $p < 0.0001$ ). The subjects who used continuous glucose monitoring per protocol ( $\geq 48$  days) improved the most ( $p < 0.0001$ ). The investigators reported that the improvement in the continuous glucose monitoring group occurred without a greater intensification of medication compared with those in the SMBG group.

Ehrhardt et al. (2010) evaluated the utility of short-term use of continuous glucose monitoring in people with type 2 diabetes on a variety of treatment modalities except prandial insulin. The investigators conducted a prospective, 52-week, two-arm, randomized trial comparing real-time continuous glucose monitoring (RT-CGM) (n = 50) versus self-monitoring of blood glucose (SMBG) (n = 50) in people with type 2 diabetes not taking prandial insulin. Real-time continuous glucose monitoring was used for four 2-week cycles (2 weeks on/1 week off). All patients were managed by their usual provider. The mean ( $\pm$  standard deviation) decline in A1C at 12 weeks was 1.0% ( $\pm$  1.1%) in the RT-CGM group and 0.5% ( $\pm$  0.8%) in the SMBG group (p = .006). There were no group differences in the net change in number or dosage of hypoglycemic medications. Those who used the RT-CGM for  $\geq$  48 days (per protocol) reduced their A1C by 1.2% ( $\pm$  1.1%) versus 0.6% ( $\pm$  1.1%) in those who used it <48 days (p = .003). Multiple regression analyses statistically adjusting for baseline A1C, an indicator for usage, and known confounders confirmed the observed differences between treatment groups were robust (p = .009). There was no improvement in weight or blood pressure. Limitations of this study included the short duration (12 weeks) for the primary endpoint (52-week results were not reported), the atypical short-term use of CGM, the fact that persons assigned to CGM tested their blood glucose more frequently than persons assigned to SMBG, and the lack of other self-care data that may explain differences between groups.

Beck et al. (2017) noted that continuous glucose monitoring (CGM) has not been well-evaluated in those with type 2 diabetes receiving insulin. The investigators reported on a manufacturer-funded randomized controlled trial to determine the effectiveness of CGM in adults with type 2 diabetes receiving multiple daily injections of insulin. The study included 158 adults who had had type 2 diabetes for a median of 17 years (interquartile range, 11 to 23 years). Participants were aged 35 to 79 years (mean, 60 years [SD, 10]), were receiving multiple daily injections of insulin, and had hemoglobin A1c (HbA1c) levels of 7.5% to 9.9% (mean, 8.5%). Subjects were randomly assigned to CGM (n = 79) or usual care (control group, n = 79). The primary outcome was HbA1c reduction at 24 weeks. The investigators reported a small but statistically significant reduction in mean HbA1c levels in the CGM group compared

to the control group. Mean HbA1c levels decreased to 7.7% in the CGM group and 8.0% in the control group at 24 weeks (adjusted difference in mean change, -0.3% [95% CI, -0.5% to 0.0%];  $P = 0.022$ ). The authors reported that the groups did not differ meaningfully in CGM-measured hypoglycemia or quality-of-life outcomes. The CGM group averaged 6.7 days (SD, 0.9) of CGM use per week. Limitations of this study included the modest reduction in HbA1c levels and limited (6-month) follow-up. In an accompanying editorial, Arguello and Freeby (2017) concluded, "With these data, we should seek to further understand patient populations that will benefit most from CGM intervention, such as those with the skills to address glucose variability. Future RT-CGM studies must also assess whether this approach improves health care outcomes for T2DM; its financial effects on the health care system; and further generalizability in T2DM subgroups, such as those with higher risk for hypoglycemia." In a commentary, Soloway et al. (2017) noted, "This study suggests that, in some patients with type 2 diabetes, CGM might lower HbA1c levels compared with standard fingerstick monitoring. However, because the study did not specify a glycemic treatment target, it doesn't prove that information from CGM — as opposed to other differences in how treating physicians managed CGM and fingerstick-monitored patients — was responsible for the lower HbA1c values. Moreover, the trial was short and set in specialty practices, and the investigators didn't assess clinical outcomes or cost-effectiveness.

There is some evidence to support the short-term diagnostic use of CGMs by medical professionals to detect unrecognized hypoglycemia, particularly unrecognized nocturnal hypoglycemia, in persons with type 2 diabetes. Chico et al. (2003) reported on the diagnostic yield of the professional CGMs in persons with type 1 ( $n = 40$ ) and type 2 diabetes ( $n = 30$ ). The investigators reported that continuous glucose monitoring detected unrecognized hypoglycemia in 62.5% of the type 1 diabetic patients and in 46.6% of the type 2 diabetic patients. The investigators noted that 73.7% of all unrecognized hypoglycemic events occurred at night. Tanenberg et al. (2004) reported on a randomized controlled trial evaluating the effect of continuous glucose monitoring in 128 subjects with insulin-treated diabetes; 10 subjects were diagnosed with type 2 diabetes. Subjects were randomly assigned to insulin

therapy adjustments based on either professional continuous glucose monitoring or self-monitoring of blood glucose values. At the end of the study, patients in both groups used the CGM for 3 days; these values were used to calculate measures of hypoglycemia. Subjects assigned to continuous glucose monitoring had a significantly shorter duration of hypoglycemia (sensor glucose less than or equal to 60 mg/dL) at week 12 of the study ( $49.4 \pm 40.8$  versus  $81.0 \pm 61.1$  minutes per event,  $p = 0.009$ ). The small number of subjects with type 2 diabetes in this study does not allow reliable conclusions to be made about the impact of continuous glucose monitoring in this subgroup. Other limitations of the study include its short duration and the use of an outcome, duration of hypoglycemia, of uncertain clinical significance. It should be noted that, in both this study by Tannenbergs et al. (2004) and the study by Chico et al. (2003) described above, continuous glucose monitoring was reported to have no significant effect on glycemic control as measured by hemoglobin A1c.

Hay et al. (2003) reported on the incidence of hypoglycemia and hypoglycemia detected by short-term (72-hour) continuous glucose monitoring in 25 elderly persons (greater than 65 years old) with type 2 diabetes treated with a sulfonylurea who were well controlled (hemoglobin A1c less than 7.5%). Elderly patients with type 2 diabetes were recruited if their glycosylated hemoglobin (HbA1c) was less than 7.5% and if their oral hypoglycemic therapy included a sulfonylurea. Patients underwent 2 consecutive 72-hour periods of continuous glucose monitoring at baseline and then again at 1 month. Patients were asked to record 4 self-monitored capillary blood glucose levels each day for calibration of the monitor and also to record meal times, exercise, and symptoms of hypoglycemia. The number of hyperglycemic (greater than 144 mg/dL), hypoglycemic (less than 50 mg/dL), and borderline-hypoglycemic (50 to 65 mg/dL) events were determined (an event was defined as a glucose value that persisted for at least 15 minutes with or without symptoms). Twenty-five patients (21 men, 4 women) aged  $73.9 \pm 4.4$  years with an HbA1c of  $6.2 \pm 0.8\%$  were each monitored for an average of 187.57 hours. The mean glucose values were: fasting,  $139 \pm 40$  mg/dL; 2 hours post-breakfast,  $167 \pm 58$  mg/dL; 2 hours post-lunch,  $157 \pm 53$  mg/dL; and 2 hours post-dinner,  $149 \pm 49$  mg/dL. Twenty patients (80%) experienced a total of 103 hypoglycemic events (less

than 50 mg/dL), and 14 of these patients experienced 54 events where the glucose levels were less than or equal to 40 mg/dL. Twenty-four patients (96%) experienced borderline hypoglycemia (50 to 65 mg/dL) ( $n = 229$  events). Patients experienced a mean of  $0.62 \pm 0.72$  episodes of hypoglycemia (interstitial glucose less than 50 mg/dL) per day (4 to 5 episodes overall),  $0.35 \pm 0.6$  episodes per day where the interstitial glucose was less than or equal to 40 mg/dL (2 to 3 episodes overall), and  $1.37 \pm 1.22$  episodes of borderline hypoglycemia (9 to 10 episodes overall). Each episode of hypoglycemia persisted for  $78 \pm 73$  minutes, and borderline hypoglycemia for  $45 \pm 11$  minutes. Patients were hypoglycemic 3.3% of the time and borderline hypoglycemic 3.7% of the time. No episode of hypoglycemia was recorded by any patient in his or her daily diary. High post-prandial glucose values (greater than 144 mg/dL 2 hours post-prandial) were recorded after 57% of all meals (breakfast 60%, lunch 57.5%, dinner 55.2%). The CGM was generally well-tolerated, but 52% of patients could not be studied for the full 12 days of monitoring. The investigators reported that hypoglycemia and excessive post-prandial glycemic excursions are common in well-controlled patients with type 2 diabetes treated with a sulfonylurea with or without metformin. Limitations of the study included the fact that it was limited to persons on oral hypoglycemics and that the study did not evaluate the impact of continuous glucose monitoring on improvements in clinical outcomes.

Zick et al. (2007) reported on the yield of 72-hour continuous glucose monitoring versus self-monitoring of blood glucose in detecting hypoglycemia (less than or equal to 60 mg/dL) in patients with type 2 diabetes on multiple daily injections of insulin. Study subjects received NPH insulin (2-week run-in) followed by insulin glargine (8-week treatment phase). Glucose levels were measured by continuous glucose monitoring and self-monitored blood glucose profiles over the 72-hour pre- and post-treatment phase. Of 367 patients in the data set, 209 patients (56.9%) experienced hypoglycemia according to continuous glucose monitoring; 97 (26.4%) recorded hypoglycemia by conventional methods. Continuous glucose monitoring and self-monitoring of blood glucose reported similar mean daytime glucose levels at baseline and



endpoint; however, nocturnal glucose levels were significantly lower with continuous glucose monitoring versus self-monitoring of blood glucose at baseline (130.2 versus 145.0 mg/dL) and at endpoint (123.3 versus 137.3 mg/dL).

The American Diabetes Association (2007) concluded that there is insufficient evidence to support the use of CGM in the hospital setting: "The introduction of real-time blood glucose monitoring as a tool for outpatient diabetes management has potential benefit for the inpatient population. However, at this time, data are lacking examining this new technology in the acutely ill patient population. Until more studies are published, it is premature to use continuous blood glucose monitoring except in a research setting."

A major limitation of CGM is the durability and stability of the glucose sensors. Interstitial glucose concentrations obtained with subcutaneous sensors correlate with blood glucose concentrations. However, the sensors become progressively less accurate over time, so they cannot be used on a maintenance basis and must be changed every 3 days. Another potential concern is the 6 to 10 minutes delay in interstitial glucose sensor response to changes in serum glucose levels. This delay appears to be most important when glucose levels are falling rapidly, since it might result in the development of clinically significant hypoglycemia before it is reflected in the sensor reading.

The Dexcom G7 Continuous Glucose Monitoring System entails a small wearable that sends real-time glucose readings to the Dexcom G7 app or receiver every 5 minutes; no finger sticks or scanning is needed. It has various features to facilitate better diabetes management decisions. The "Urgent low soon alert" setting provides a 20-minute advance warning of when one's glucose level will reach 55 mg/dL so that one can act quickly and avoid a potential hypoglycemic event. In the "Silence all" setting, all alerts will not sound or vibrate for up to 6 hours. Visual notifications will still appear on the lock screen. The "Delay 1st alert" setting

allows one to delay one's "First high alert" until one's sensor reading is high for a while (user chooses how long). The "Quiet mode" setting allows all users' G7 alerts to vibrate; users' "Urgent low glucose and technical alerts" will still escalate to sound if not acknowledged.

Garg et al. (2022) examined the safety and accuracy of a 7th generation (G7) Dexcom CGM during 10.5 days of use in adults with diabetes. Adults with either T1 or T2 DM (on intensive insulin therapy or not) participated at 12 investigational sites in the U.S. In-clinic visits were carried out on days 1 or 2, 4 or 7, and on the 2nd half of day 10 or the 1st half of day 11 for frequent comparisons with comparator blood glucose measurements obtained with the venous blood glucose measurements (YSI) 2300 Stat Plus glucose analyzer. Participants wore sensors concurrently on the upper arm and abdomen. Accuracy evaluation included the proportion of CGM values within 15% of comparator glucose levels of greater than 100 mg/dL or within 15 mg/dL of comparator levels of 100 or less mg/dL (%15/15), along with the %20/20 and %30/30 agreement rates. The mean absolute relative difference (MARD) between temporally matched CGM and comparator values was also calculated. Data from 316 participants (619 sensors, 77,774 matched pairs) were analyzed. For arm- and abdomen-placed sensors, overall MARDs were 8.2% and 9.1%, respectively. Overall %15/15, %20/20, and %30/30 agreement rates were 89.6%, 95.3%, and 98.8% for arm-placed sensors and were 85.5%, 93.2%, and 98.1% for abdomen-placed sensors. Across days of wear, glucose concentration ranges, and rates of change, %20/20 agreement rates varied by no more than 9% from the overall %20/20. No serious AEs were observed. The authors concluded that the G7 CGM provided accurate glucose readings with single-digit MARD with arm or abdomen placement in adults with DM.

Laffel et al. (2023) examined the accuracy of a 7th-generation "G7" CGM system in children and adolescents with T1DM. Sensors were worn on the upper arm and abdomen. The CGM data were available from 127 of 132 participants, aged 7 to 17 years, across 10.5 days of use, various glucose concentration ranges, and various rates of glucose change for comparisons with temporally matched YSI. Data were also available from 28 of 32 participants, aged 2 to 6

years, for whom capillary (finger stick) blood provided comparator glucose values. Accuracy metrics included the mean absolute relative difference (MARD) between CGM and comparator glucose pairs, the proportion of CGM values within 15 mg/dL or 15% of comparator values of less than 100 or 100 or greater than mg/dL, respectively, and the analogous %20/20 and %30/30 agreement rates. For participants aged 7 to 17 years, a total of 15,437 matched pairs were obtained from 122 arm-placed and 118 abdomen-placed sensors. For arm-placed sensors, the overall MARD was 8.1%, and overall %15/15, %20/20, and %30/30 agreement rates were 88.8%, 95.3%, and 98.7%, respectively. For abdomen-placed sensors, the overall MARD was 9.0%, and overall %15/15, %20/20, and %30/30 agreement rates were 86.0%, 92.9%, and 97.7%, respectively. Good accuracy was maintained across wear days, glucose ranges, and rates of glucose change. Among those aged 2 to 6 years, a total of 343 matched pairs provided an overall MARD of 9.3% and an overall %20/20 agreement rate of 91.5%. The authors concluded that the G7 CGM placed on the arm or abdomen was accurate in children and adolescents with T1DM.

A draft evidence review prepared by the Center for Evidence-Based Policy for the Washington State Health Care Authority Health Technology Assessment Program (Durbin et al., 2025) found that CGM use in adults with type 2 diabetes on non-intensive insulin regimens resulted in small reductions in HbA1c from baseline that were statistically significant but did not meet the prespecified threshold for clinical significance. The evidence review identified seven randomized controlled trials in 15 publications (N = 802; follow-up range, 12 to 52 weeks) that assessed CGM among adults with type 2 diabetes on non-intensive insulin regimens (1 to 3 injections per day). The investigators found that CGM use resulted in a small reduction in HbA1c from baseline that was statistically significant (moderate certainty of evidence, based on seven RCTs). At final follow-up (range, 12 to 52 weeks), CGM use was associated with a significant reduction in HbA1c compared with no CGM (pooled mean difference, -0.27%; 95% CI, -0.46 to -0.08; p = .005). However, this difference did not meet the threshold for clinical significance (minimal clinically important difference, 0.5% change). There was no difference between CGM and self-monitoring of blood glucose (SMBG) groups in the proportion of

participants who achieved target HbA1c levels (i.e., 7.0%, 7.5%) at 12 or 24 weeks (low certainty of evidence, based on one RCT). There was no clear association between CGM and improved diabetes-related quality of life (QoL) (low certainty of evidence, based on four RCTs). There was no clear association between CGM and improved general QoL (low certainty of evidence, based on two RCTs). The evidence review did not identify any eligible randomized controlled trials evaluating the effectiveness or safety of CGMs in children with type 2 diabetes on non-intensive insulin regimens.

The evidence review found that, in adults with type 2 diabetes on oral hypoglycemic medications, there was no consistent difference in change in HbA1c from baseline with CGM versus other non-CGM monitoring methods (low certainty of evidence, based on six RCTs) (Durbin et al., 2025). They did not identify any eligible randomized controlled trials evaluating CGMs in adults with type 2 diabetes not on insulin or oral hypoglycemic medication regimens. In adults with type 2 diabetes on mixed non-intensive hypoglycemic therapies, there was no consistent difference in change in HbA1c from baseline with CGM versus other non-CGM monitoring methods (very low certainty of evidence, based on five RCTs).

### *Continuous Glucose Monitoring for Individuals with Type 1a Glycogen Storage Disease*

HersHKovitz et al. (2001) noted that glycogen storage disease type I (GSD I) is characterized by impaired production of glucose from glycogenolysis and gluconeogenesis, resulting in severe fasting hypoglycemia. These investigators examined the efficacy of a continuous glucose monitoring system (CGMS) (MiniMed) to determine the magnitude and significance of hypoglycemia in GSD I and assessed the efficacy of its dietary treatment. A total of four children with GSD I were studied over a 72-hour period. Results indicated that the values recorded with a subcutaneous CGMS were highly correlated with paired blood glucose values measured by glucometer. Significant periods of asymptomatic hypoglycemia were noted,

especially during nighttime. The authors concluded that the findings of this study suggested that repeated continuous subcutaneous glucose monitoring may serve as a useful tool for the assessment of the long-term management of GSD I patients.

Maran et al. (2004) stated that the development of new systems for CGM has recently increased interest in their potential applications among physicians involved in diabetes care. One of the most common applications of such devices is the identification of hypoglycemic events in insulin-treated diabetic patients (particularly during the night) and the evaluation of the full daily glucose excursions. Among commercially available glucose sensors, the Glucoday System has been used for practical clinical application in the last two years. One of the most important features of this device is its accuracy in monitoring interstitial glucose values, specifically in the hypoglycemic range. This feature is clinically relevant when applied in the clinical setting of patients with T1DM. The ability to monitor glucose continuously could indeed be a useful tool for the study of hypoglycemic conditions other than diabetes. In patients with hyperinsulinemic hypoglycemia, recurrent episodes of asymptomatic hypoglycemia are common, and in patients with GSDs, avoidance of recurrent and prolonged hypoglycemic episodes usually requires frequent determinations by means of home blood glucose monitoring. The authors concluded that experimental preliminary evidence suggested that this new technology could be applied in the clinical setting to help the physician identify mainly nocturnal hypoglycemic events, otherwise not revealed by traditional self-blood glucose monitoring, even in those patients who are not treated by conventional insulin therapy.

White and Jones (2011) noted that CGMS are now in widespread use in diabetes management with an increasing evidence base; nevertheless, there are few reports of their use in GSD. Liver GSDs are most often managed by intensive dietary regimens; however, risks of over- and under-treatment remain. These researchers described their use of CGMS in a cohort of GSD patients, the results obtained, and the frequency of complications. Their experience was that CGM is a reliable, well-accepted, and valid tool in the monitoring of GSD patients and allows for assessment of blood sugar control in the "real-life" setting, unlike

hospital admissions. Combining CGM with urine ketone and/or blood lactate measurements, again at home, improves the investigation yet further. It is possible to perform CGM for periods including both school days and weekends, and also to change the dietary regimen during the period of monitoring to reduce the frequency of assessments. Risks of decreased reliability in the low range of blood sugars may be outweighed by the increased validity of the patient being in the home environment, with a normal diet and activity schedule.

Kasapkara et al. (2014) noted that GSD I is an autosomal recessive metabolic disorder caused by defects in the glucose-6-phosphatase complex. Deficient activity in the glucose-6-phosphatase- $\alpha$  catalytic unit characterizes GSD Ia, and defects in the glucose-6-phosphate transporter protein characterize GSD Ib. Type Ia involves the liver, kidney, and intestine (and Ib also leukocytes), and the clinical manifestations are hepatomegaly, failure to thrive, severe fasting hypoglycemia within 3 to 4 hours after a meal, hyperlactatemia, hyperuricemia, and hyperlipidemia. These researchers examined the safety and efficacy of a subcutaneous CGMS to determine the magnitude and significance of hypoglycemia in GSD I and evaluated the efficacy of the revised dietary treatment. A total of 16 children with GSD I were studied over a 72-hour period; CGM was repeated in all patients 3 to 6 months after the first monitoring to examine the effects of revised dietary instructions on glycemic control. All the patients completed the study without any major adverse events (AEs). Significant periods of asymptomatic hypoglycemia (below 4 mmol/L, 70 mg/dL) were noted. There was a close correlation between CGM sensor and capillary blood glucose values measured by a glucometer. CGM indicated a considerable reduction in the duration of hypoglycemia, liver size, and improvements in secondary metabolic derangements such as hyperlactacidemia and hyperlipidemia. The authors concluded that CGM could be applied in the clinical setting to help the physician identify hypoglycemic events, and repeated CGM may serve as a safe and useful tool for the assessment of the long-term management of patients with GSD I; however, the long-term benefits on metabolic control remain to be determined.